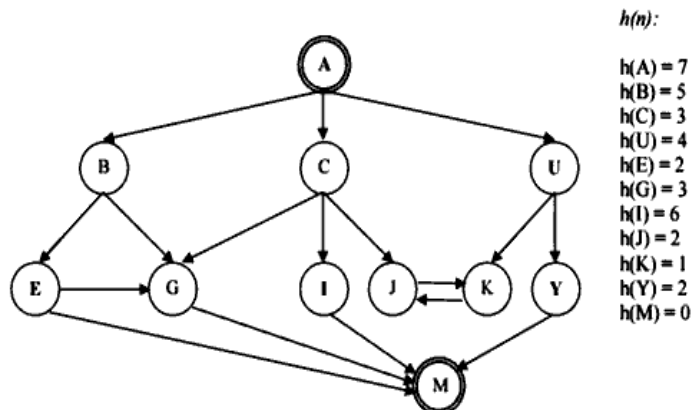


Instructions to the Students:

1. All the questions are compulsory.
2. The level of question/expected answer as per OBE or the Course Outcome (CO) on which the question is based is mentioned in () in front of the question.
3. Use of non-programmable scientific calculators is allowed.
4. Assume suitable data wherever necessary and mention it clearly.

	(Level/CO)	Marks
Q.1 Solve Any Two of the following.		12
A) What do you mean by AI? Explain contribution of AI in various fields.	CO1	6
B) What is Intelligent Agent? Explain structure of agent and an example with PEAS Property for Automatic Taxi Driving.	CO1	6
C) Explain the difference between:	CO2	6
i. Deductive and Inductive learning		
ii. Forward and Backward reasoning		
Q.2 Solve Any Two of the following.		12
A) Express the following as Semantic net with interconnected nodes and labeled arcs.	CO7	6
“Company ABC is software development company. Three departments within the company are Sales, Administration and Programming. Joe is Manager of Programming. Bill and Sue are programmer. Sue is married to Sam. Sam is editor of PHI. They have three children and they live on Elm Street. Sue wears glasses and is 5.5 feet tall. ”		
B) Find the search steepest Ascent hill climbing for the following graph.	CO2	6



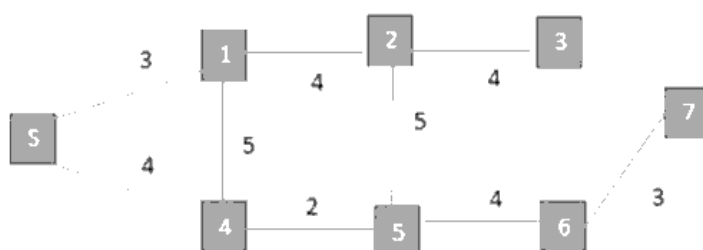
- C) i. Explain State-Space Search technique in short. CO2 6
 ii. Solve following Crypt Arithmetic Problem SEND + MORE.

Q.3 Solve Any Two of the following. 12

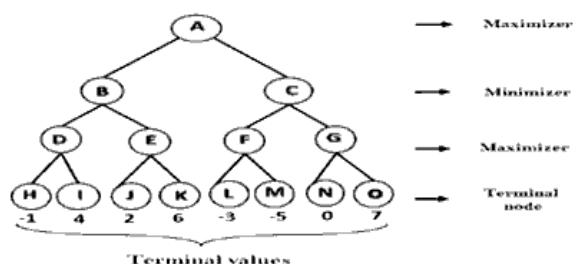
- A) Explain Constraint Satisfaction Problem with an example of graph coloring. CO3 6

- B) Consider the graph given in figure below. Assume that the-initial state is S and the goal state is 7. Find a path from the initial state to the goal state using A* Search. Also report the solution cost. The straight line distance heuristic estimates for the nodes are as follows:

$$h(1)=14, h(2)=10, h(3)=8, h(4)=12, h(5)=10, h(6)=10, h(s)=16$$



- C) Explain mini-max search procedure and solve the problem given below. CO4 6



Q.4 Solve Any Two of the following. 12

- A) What is Quantifiers in First-order Logic? Compare and Contrast Universal quantifier and Existential quantifier with suitable example. CO6 6

- B) Represent the following statements in predicate logic: CO6 6

- Marcus tried to cheat Caesar.
- All Pompeian's were Roman.
- All Romans were either loyal to Caesar or hated him.
- Everyone is loyal to someone.
- People only try to co-operate rulers they are not loyal to.

- C) What are the Heuristics? What is its importance? Describe its types with the help of examples. Also justify the sentence "Heuristics are not sure to lead to a situation yet the field of AI is full of them." CO3 6

Q. 5	Solve Any Two of the following.		12
A)	Define Planning. What are the different components of planning? Explain in detail goal stack planning.	CO5	6
B)	Define Expert system. Explain the architecture of expert system in detail with a neat diagram and an example.	CO7	6
C)	Explain the applications of Natural Language Processing	CO6	6

***** END OF QUESTION PAPER*****